

Appendix D

On the Scan Versus the Embedded Impedance

D.1 INTRODUCTION

When working with phased arrays, it is of particular interest to compare the scan impedance with the embedded impedance, in particular whether one can be derived from the other.

The first of these, namely the scan impedance, is the impedance observed at the element terminals when the proper voltages are applied to *all* the elements. These voltages may all have the same amplitude corresponding to uniform aperture illumination or they may be tapered across the aperture. When we vary the phases of these terminal voltages in a linear fashion across the aperture, the beam will scan in different directions and the scan impedance will in general vary considerably with scan angle.

The embedded impedance, on the other hand, is the terminal impedance observed at just one element usually located somewhere in the middle of the array and with all the other elements terminated in loads (usually resistive). Thus, a terminal voltage is only applied to a single element, while all the other elements are excited parasitically.

It is also of interest to consider the case where we excite a single column or row with identical voltages while all the other elements are terminated in loads like before. The impedance measured at the terminals of the column of elements in question is often called the embedded stick impedance $Z_{\text{emb stk}}$. We shall see that it often makes more sense to consider the embedded stick impedance rather than the embedded impedance of a single element.

Although there are similarities between the scan and embedded impedances, they are distinctly different. Only the scan impedance can be associated with the direction of the beam, while the embedded impedance can be associated with only one direction. (Actually the pattern looks more like an element pattern with ripples.) However, not even for broadside scan do we obtain more than an occasional similarity between the two types of impedance.

Needless to say, the embedded impedance for a single element is by far the simplest of the three types of impedance to implement. You only need a single connector at one element in the middle of the array, while all the others are terminated by resistors that can be either soldered in place or simply printed. In contrast, measurements of the scan impedance may require a connector or equivalent at each element, and, most importantly, we must be able to apply voltages at each terminal and be able to control these voltages in phase as well as amplitude.

For these reasons there has long been a strong tendency to measure just the embedded impedance of a single element. While this may be a good way to check certain fundamental features of the array and in some cases the presence of surface waves, it cannot be emphasized enough that it is *not* a substitute for measuring the scan impedance, not even at broadside. And of these two impedances, the latter is by far the most important. In fact the embedded impedance has always reminded the author of an honorable degree. There is not much you can do with it except marvel at it!

But let us now look at the facts of the scan and the embedded impedances.

D.2 THE SCAN IMPEDANCE

The scan impedance has been discussed extensively in reference 135. Thus, only the highlights will be given here for easy reference.

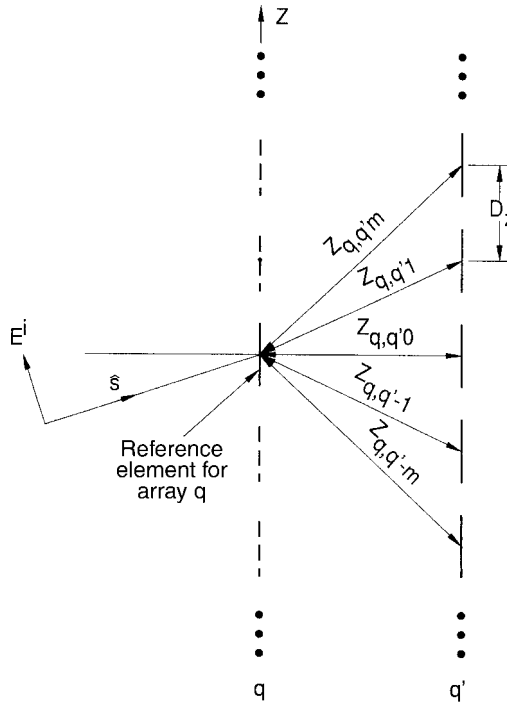
Consider Fig. D.1, where we show two infinite stick arrays q and q' . We denote the mutual impedance between the reference element in array q and element m in array q' by $Z_{q,q'm}$. The mutual impedance between the reference element in array q and all the other elements in array q' is then defined as

$$Z^{q,q'} = \sum_{m=-\infty}^{\infty} Z_{q,q'm} e^{-j\beta m D_z s_z}. \quad (\text{D.1})$$

Let us now consider a finite $\times$ infinite array as shown in Fig. D.2 consisting of $2Q + 1$ stick arrays. The current in the reference elements are denoted I_q . Writing Ohm's Law for the reference element in stick array number 0 yields

$$V^{0,0} = \sum_{q=-Q}^Q Z^{0,q} I_q. \quad (\text{D.2})$$

We now let $Q \rightarrow \infty$ such that the array becomes infinite $\times$ infinite. In that event the array is truly periodic in the X direction as well as in the Z direction and



$$Z^{q,q'} = \sum_{m=-\infty}^{\infty} Z_{q,q'm} e^{-j\beta m D_z s_z}$$

Fig. D.1 Evaluation of the array mutual impedance $Z^{q,q'}$ between the reference element in array q and all the other elements in array q' by using the mutual impedance approach.

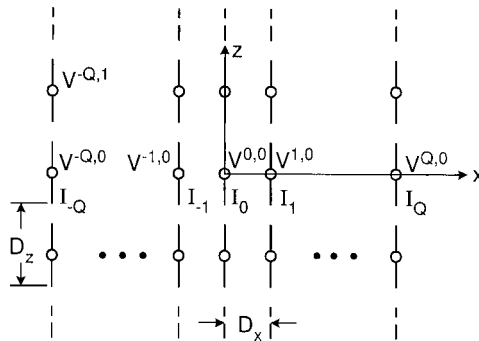


Fig. D.2 A finite $\times$ infinite array comprised of $2Q + 1$ infinitely long stick arrays. All elements are driven with voltage generators $V^{q,m}$, and the currents in the reference elements in row 0 are denoted I_q . For $Q \rightarrow \infty$ we obtain an infinite $\times$ infinite array where Floquet's Theorem yields $I_q = I_0 e^{-j\beta q D_x s_x}$.

Floquet's Theorem applies [26], yielding

$$I_q = I_0 e^{-j\beta q D_x s_x}. \quad (\text{D.3})$$

Substituting (D.3) into (D.2), we obtain

$$V^{0,0} = I_0 \sum_{q=-\infty}^{\infty} Z^{0,q} e^{-j\beta q D_x s_x}. \quad (\text{D.4})$$

Thus, the scan impedance for an infinite array without a groundplane is

$$Z_A = \frac{V^{0,0}}{I_0} = Z^{0,0} + 2 \sum_{q=1}^{\infty} Z^{0,q} \cos(\beta q D_x s_x). \quad (\text{D.5})$$

A plot of the scan impedance Z_A for an infinite $\times$ infinite array without a groundplane and with the same interelement spacings as used in Chapter 6 as obtained from the PMM code is shown in Fig. D.3. The impedance is plotted in a Smith chart normalized to 100 ohms. Furthermore, the impedance in Fig. D.3 as well as in later figures includes the matching of a short transmission line with characteristic impedance 200 ohms and length 0.13 cm. This was typically true in the cases shown in Chapter 6, which should make comparisons between the various curves more meaningful. We remind the reader that the purpose of this transmission line (also referred to as a "pigtail") was to better center the impedances as well as compress them (for details see Chapter 6 and Appendix B).

D.3 THE EMBEDDED STICK IMPEDANCE

The embedded stick impedance is perhaps best illustrated by considering just three stick arrays as shown in Fig. D.4. Each element in the center column is fed by voltage generators with voltages $V^{0,m}$, where m refers to the row number. The two outer stick arrays are not fed but only loaded with identical load impedances Z_L .

By writing Ohm's Law for each reference element in row 0, we obtain

$$\begin{bmatrix} 0 \\ V^{0,0} \\ 0 \end{bmatrix} = \begin{bmatrix} Z^{-1,-1} + Z_L & Z^{-1,0} & Z^{-1,1} \\ Z^{0,-1} & Z^{0,0} & Z^{0,1} \\ Z^{1,-1} & Z^{1,0} & Z^{1,1} + Z_L \end{bmatrix} \begin{bmatrix} I_{-1} \\ I_0 \\ I_1 \end{bmatrix}. \quad (\text{D.6})$$

Applying Cramer's rule, we obtain from (D.6)

$$I_0 = \frac{1}{D} \begin{vmatrix} Z^{-1,-1} + Z_L & 0 & Z^{-1,1} \\ Z^{0,-1} & V^{0,0} & Z^{0,1} \\ Z^{1,-1} & 0 & Z^{1,1} + Z_L \end{vmatrix}, \quad (\text{D.7})$$

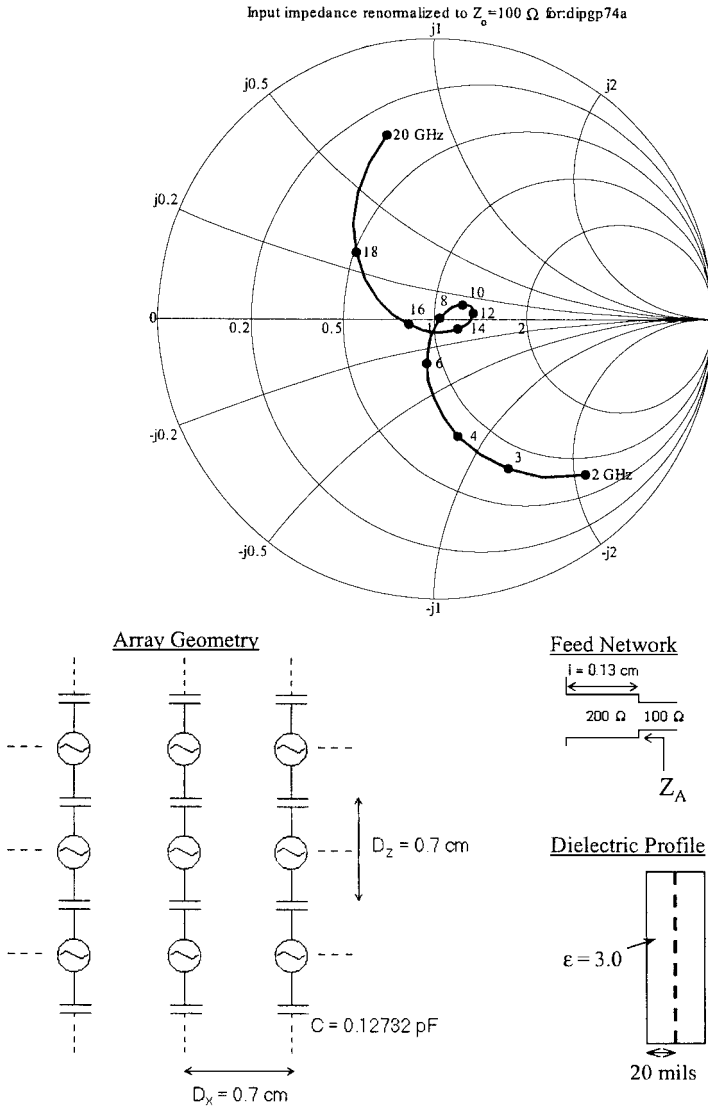


Fig. D.3 The scan impedance Z_A for an infinite $\times$ infinite array without a groundplane obtained from the PMM program. The array dimensions are identical to the broadband array in Chapter 6 (see insert) but includes only the “underwear” (see “Dielectric Profile” above). Also included is a small transmission line matching section as was the case in Chapter 6 (see text).

where D denotes the determinant of (D.6). Expanding (D.7) along the second column yields

$$I_0 = \frac{V^{0,0}}{D} [(Z^{-1,-1} + Z_L)(Z^{1,1} + Z_L) - Z^{-1,1}Z^{1,-1}]. \quad (\text{D.8})$$

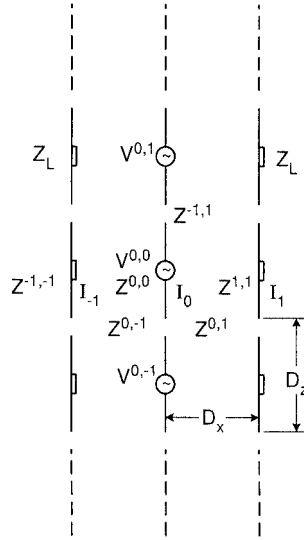


Fig. D.4 The embedded stick impedance $Z_{\text{emb stk}}$ is the terminal impedance observed in the center stick array when all terminals in the center array are fed with voltages $V^{0,m}$ and all other stick arrays are loaded with identical load impedances and merely excited parasitically.

Furthermore, we obtain for the determinant D of (D.6) by expanding along the second column

$$\begin{aligned} D = & -Z^{-1,0}[Z^{0,-1}(Z^{1,1} + Z_L) - Z^{0,1}Z^{1,-1}] \\ & + Z^{0,0}[(Z^{-1,-1} + Z_L)(Z^{1,1} + Z_L) - Z^{-1,1}Z^{1,-1}] \\ & - Z^{1,0}[(Z^{-1,-1} + Z_L)Z^{0,1} - Z^{-1,1}Z^{0,-1}]. \end{aligned} \quad (\text{D.9})$$

Substituting (D.9) into (D.8) finally yields for the embedded stick impedance

$$\begin{aligned} Z_{\text{emb stk}} &= \frac{V^{0,0}}{I_0} \\ &= -Z^{-1,0} \frac{Z^{0,-1}(Z^{1,1} + Z_L) - Z^{0,1}Z^{1,-1}}{(Z^{-1,-1} + Z_L)(Z^{1,1} + Z_L) - Z^{-1,1}Z^{1,-1}} + Z^{0,0} \\ &\quad - Z^{1,0} \frac{(Z^{-1,-1} + Z_L)Z^{0,1} - Z^{-1,1}Z^{0,-1}}{(Z^{-1,-1} + Z_L)(Z^{1,1} + Z_L) - Z^{-1,1}Z^{1,-1}}. \end{aligned} \quad (\text{D.10})$$

Inspection of (D.10) shows that the embedded stick impedance $Z_{\text{emb stk}}$ consists of the stick self impedance $Z^{0,0}$ for the center array plus two terms representing the overcoupled impedances from the two outer arrays.

It can be shown that the embedded stick impedance for any size finite array is structured the same way—that is, as a sum of the stick self-impedance $Z^{0,0}$ plus overcoupled terms associated with all the other parasitically excited stick arrays.

D.3.1 Example

We have calculated the stick self-impedance $Z^{0,0}$ for an array without ground-plane with dimensions as used in Chapter 6 as obtained from the SPLAT program. This program cannot handle dielectric slabs; however, it will approximate the effect of the dielectric “underwear” [136] by placing cylindrical dielectric shells around each element. The thickness of these dielectric coatings should be approximately equal to the thickness of the “underwear.”

Thus, we show in Fig. D.5 an example of the stick self-impedance $Z^{0,0}$ with dimensions as shown in the insert and as obtained from the SPLAT program (includes a matching transmission line).

Furthermore, we show in Fig. D.6 the embedded stick impedance $Z_{\text{emb stk}}$ as given by (D.10) as also obtained from the SPLAT program and with dimensions as given in the insert (includes matching transmission line).

Comparing $Z^{0,0}$ in Fig. D.5 and $Z_{\text{emb stk}}$ in Fig. D.6 shows that they are in fact very similar except that the latter has a larger loop at the middle frequencies as caused by the overcoupled terms observed in (D.10). At the lower and upper frequencies, $Z^{0,0}$ and $Z_{\text{emb stk}}$ are practically identical. More comments about this subject will be given in the conclusion section. [A parametric study (not shown) reveals that the impedance loop of $Z_{\text{emb stk}}$ becomes larger for smaller values of Z_L and vice versa. See also comments in the Postscript section of this chapter.]

D.4 THE EMBEDDED ELEMENT STICK IMPEDANCE

In the previous section we considered the impedance properties when an entire column array was fed at all its terminals. However, as was pointed out already in the Introduction, the embedded impedance will in general be obtained by exciting only one pair of terminals while all other terminals are just loaded.

It is therefore of interest to investigate just a single stick array when we feed only a single pair of terminals while the rest are loaded with the same load impedances Z_L . We have denoted the terminal impedance for this case for the embedded element stick impedance $Z_{\text{emb ele stk}}$. Examples are shown in Fig. D.7. The array has the same dimensions as used in the previous section (see insert). The calculations were obtained from the method of moment program ESP [137]. Similar to the SPLAT program used to obtain the results in Figs. D.5 and D.6, it uses dielectric cylinders placed around each element.

Finally we show in Fig. D.8 the self-impedance of a single dipole without capacitors or any other elements present but includes matching sections (see text earlier). As we would expect, the real part is low and the reactance is capacitive and high. Obviously, this is what sticks in peoples’ minds when they say that the type of array discussed in Chapter 6 will never have a chance for yielding a broad bandwidth (no pun intended). Note how the impedance “shrinks” as more elements are added and excited as observed in Fig. D.7.

It is of course the mutual coupling that comes to our rescue. We will elaborate on this subject in the next section.

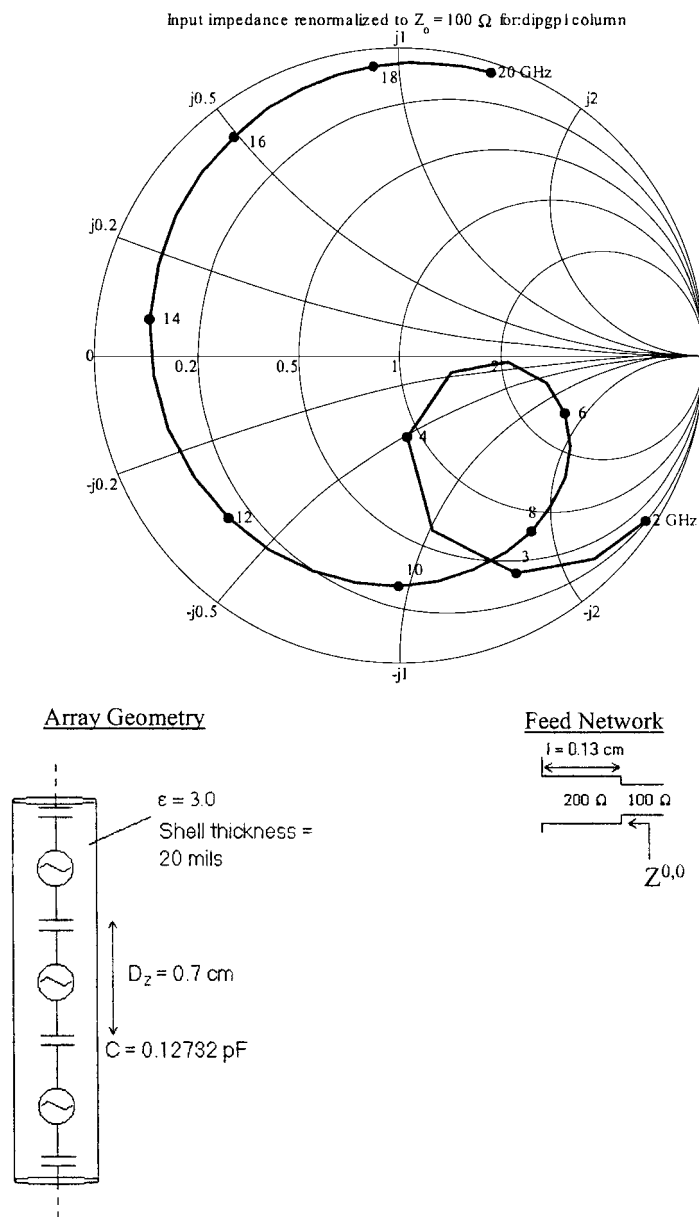


Fig. D.5 The stick self-impedance $Z^{0,0}$ for an array without a groundplane. Element dimensions identical to the case in Fig. D.3 (see insert). Obtained from the SPLAT program. The dielectric “underwear” is modeled by placing dielectric shells around the elements. Diameter approximately equal to total thickness of the underwear. Includes a matching transformer (see text).

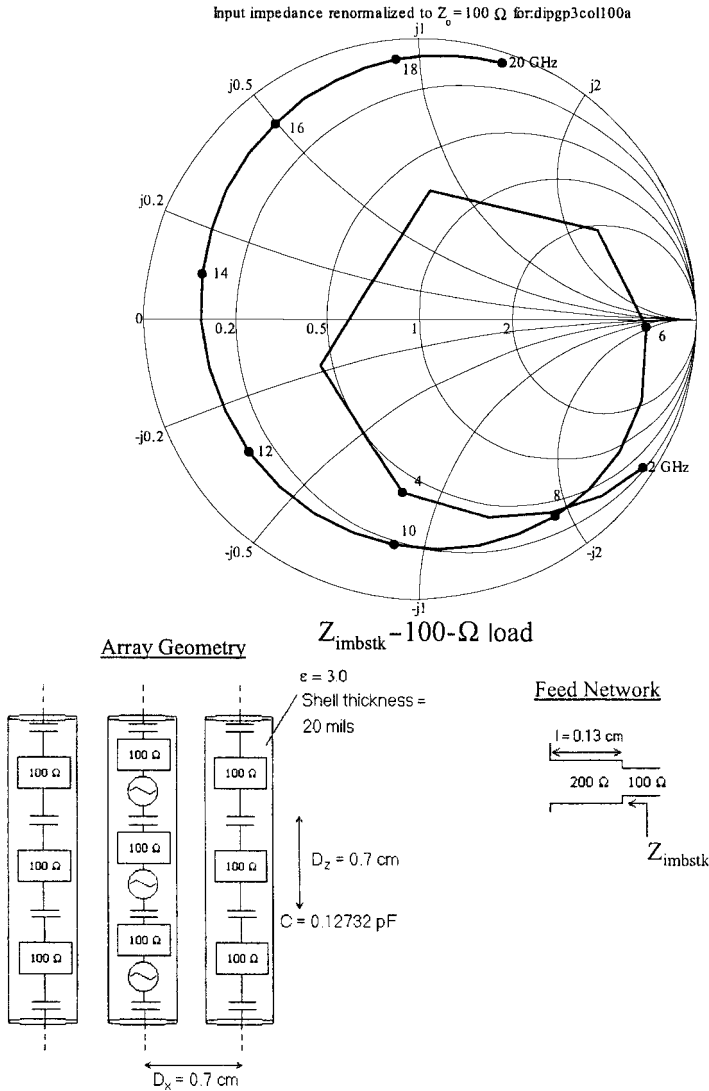


Fig. D.6 The embedded stick impedance $Z_{emb \text{ stk}}$ as given by (D.10) (no groundplane). The center stick array is driven while the two outer stick arrays are just loaded with $Z_L = 100 \text{ ohms}$. Array dimensions as in Figs. D.3 and D.4 (see insert). From the SPLAT program. Includes matching transformer (see text).

D.5 ON THE SCAN IMPEDANCE OF A FINITE ARRAY

In the previous sections we observed that the effect of the parasitic columns was practically nil at the lower and upper frequencies while it had a deteriorating effect at the middle frequencies. However, we already know that the scan impedance

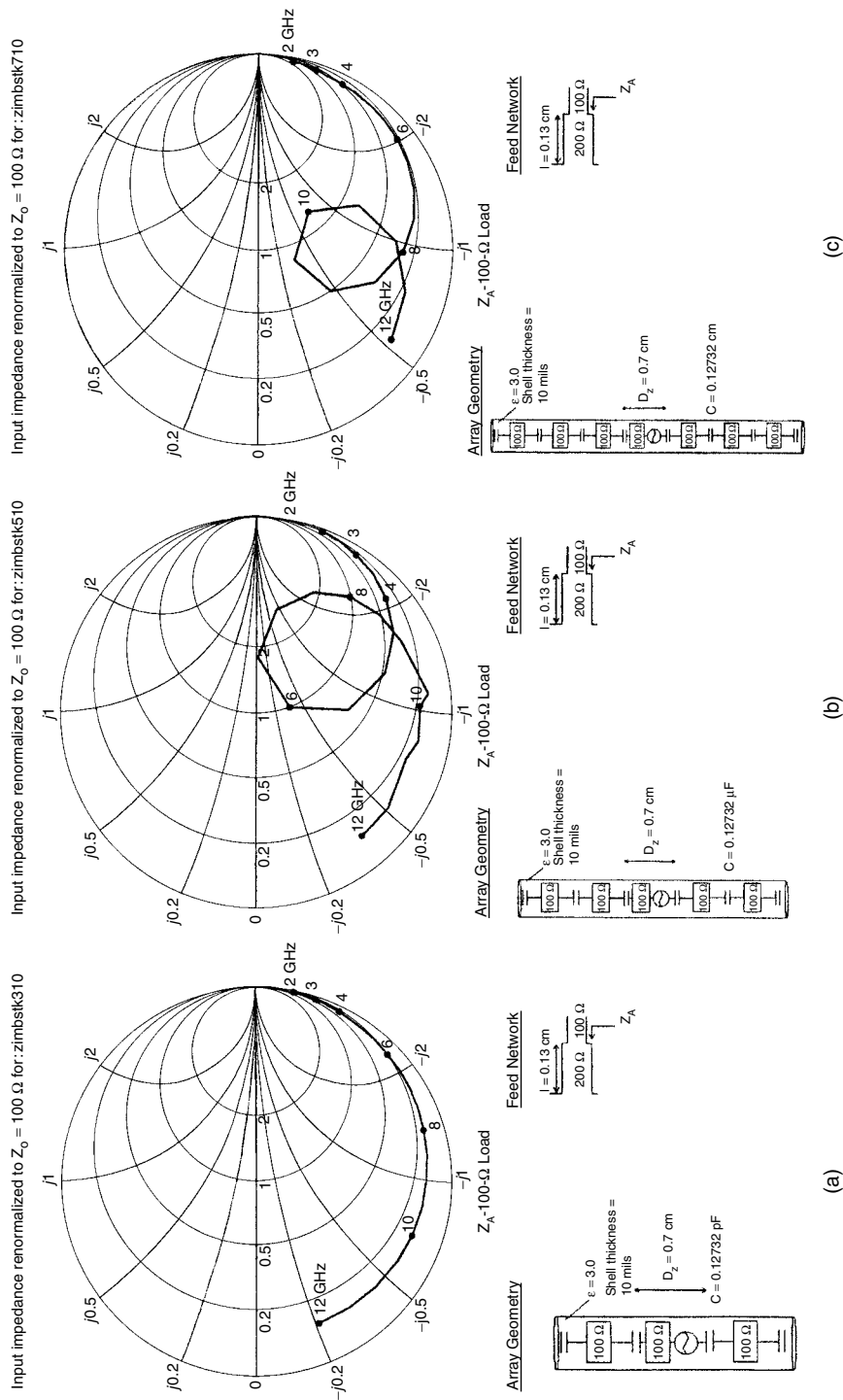


Fig. D.7 The embedded element stick impedance $Z_{\text{emb}}^{\text{ele}} \text{ sik}$ for element dimensions identical to Fig. D.6. From the ESP program, Dielectric “underwear” modeled by dielectric cylinder around the elements. (a) With one parasitically excited dipole on top and bottom. Loaded with $Z = 100 \text{ ohms}$. (b) With two parasitically excited dipoles on the top and two on the bottom. Loaded with $Z_L = 100 \text{ ohms}$. (c) With three parasitically excited dipoles on the top and bottom. Includes matching section (see text).

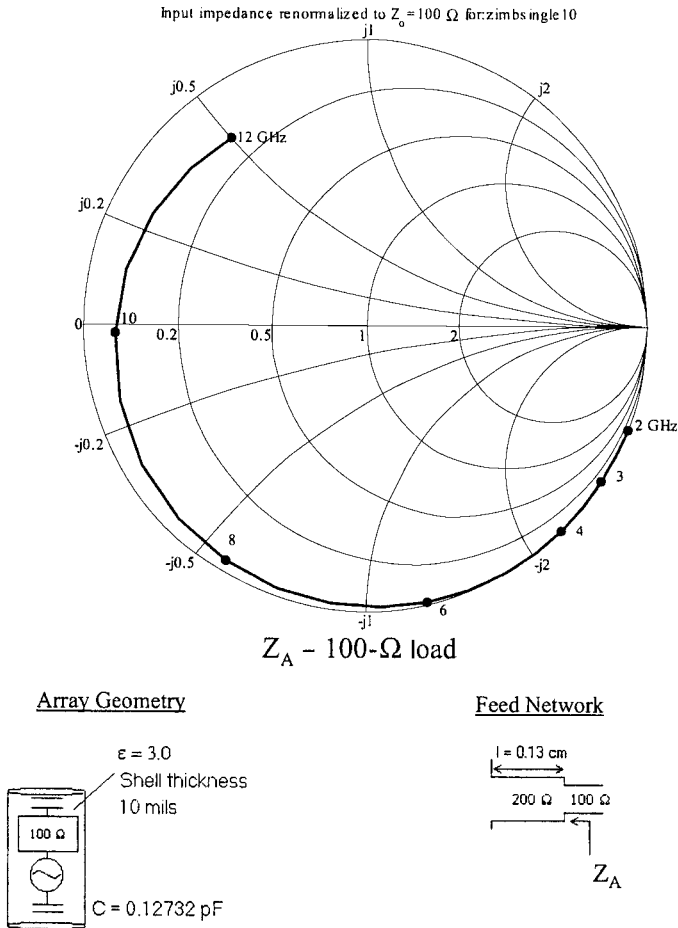


Fig. D.8 The self-impedance of a single dipole with total length equal to the element section used in Fig. D.7 (see insert). A dielectric cylinder was placed around the element to model the "underwear." Includes a matching section (see text).

Z_A for an infinite array is considerably "better" in the entire frequency range (see Fig. D.3). It is therefore pertinent to ask the question, Would it not be better to excite the parasitic columns with voltage generators like the center column rather than parasitically?

Thus, we therefore show the scan impedance for the center column in Fig. D.9 when the two outer columns are excited as well. Similarly we show the scan impedance for the two outer columns in Fig. D.10. Comparing the scan impedances in Figs. D.9 and D.10 with the embedded impedance $Z_{\text{emb stk}}$ in Fig. D.6 readily shows a significant improvement in particular at the **lower frequencies**. In the midrange we observe the presence of a surface wave. It has been suppressed by feeding each element with a voltage generator that includes a

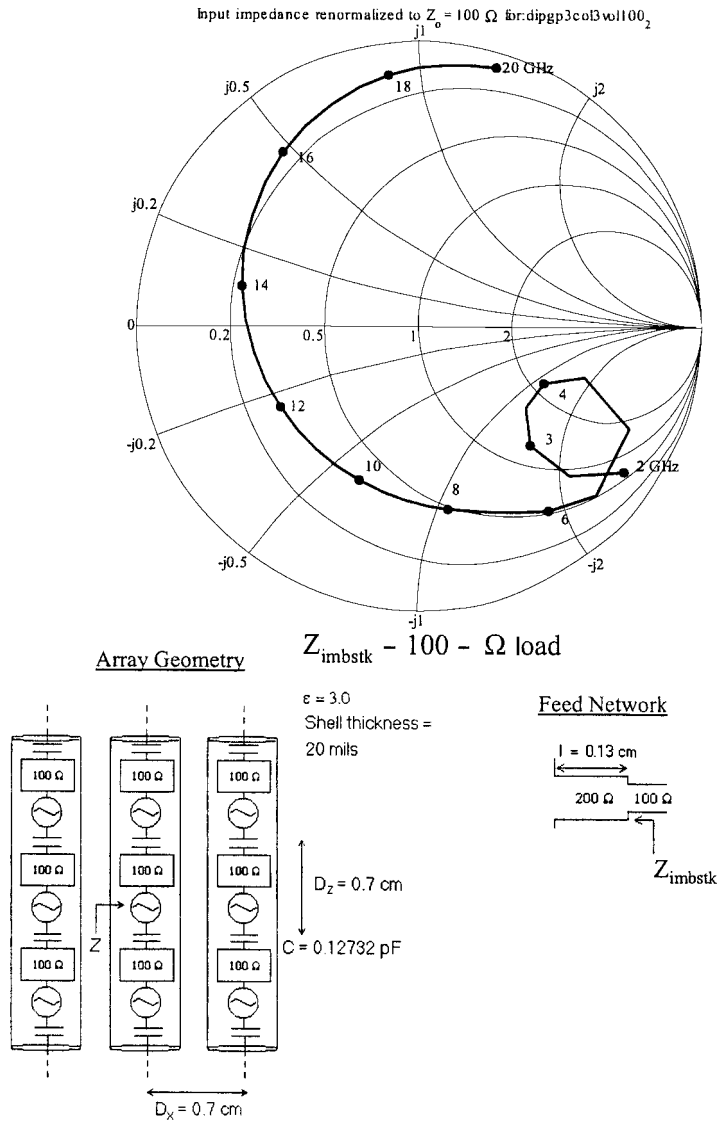


Fig. D.9 The scan impedance for the center column when all three columns are fed with identical voltage generators with generator impedances equal to 100 ohms (to suppress surface waves). Dielectric cylinders are placed around the elements to model the “underwear.” Includes matching section (see text). From the SPLAT program.

generator impedance equal to 100 ohms as discussed in detail in Chapter 4. This resistor is subtracted in the figure.

We have further verified this observation for the case when we are using five columns rather than three (not shown).

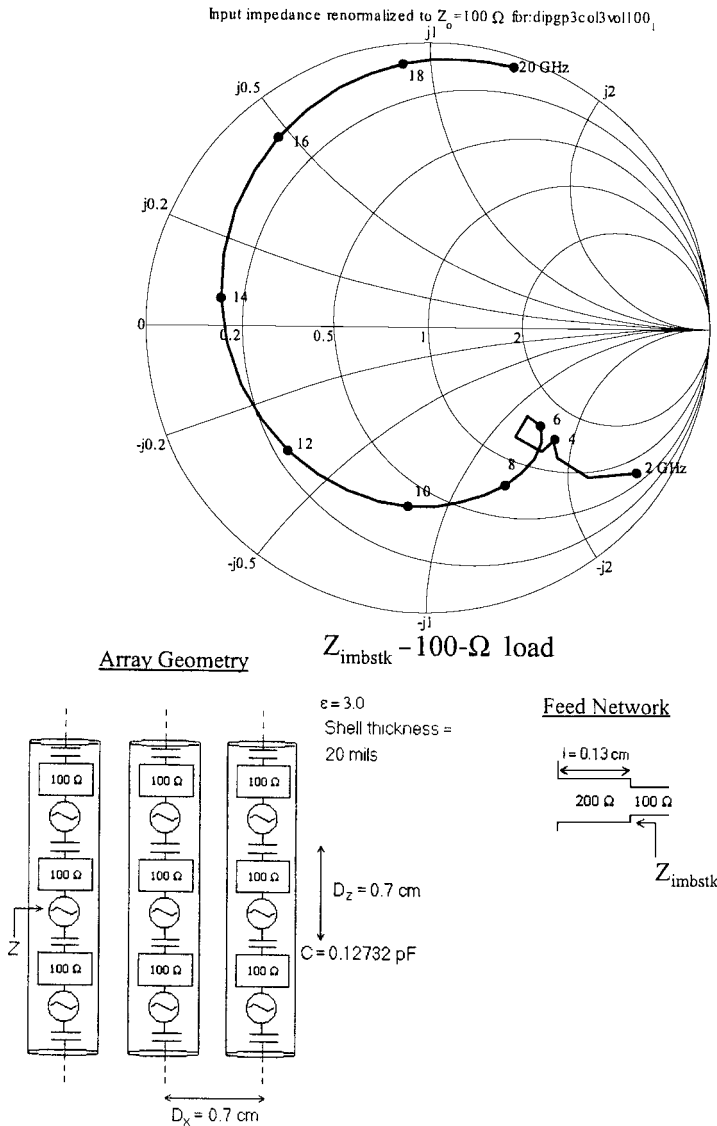


Fig. D.10 The scan impedance for the outer columns when all three columns are fed with identical voltage generators with generator impedance equal to 100 ohms to suppress surface waves. Dielectric cylinders are placed around the elements to model the "underwear." Includes matching section (see text). From the SPLAT program.

We may conclude that use of parasitic columns either is ineffective or simply makes things worse. It appears prudent to recommend to excite all columns in a finite array and not try to improve upon the impedance by garnishing the active part of the array with all kinds of dummy elements located next to them.

We finally show in Fig. D.11 the terminal impedance as seen from the center element with two parasitic sections at the top and two below like Fig. D.7b but also flanked on each side with parasitically excited columns comprised of five segments. Dimensions are given in the insert, and the ESP program was used with a 10-mil dielectric sheet. As usual, the impedance curves include the small transmission line section. Note the poor performance at the lower frequencies compared to the scan impedance in Fig. D.3.

D.6 HOW TO MEASURE THE SCAN IMPEDANCE Z_A

It does not seem possible to obtain the scan impedance Z_A from any embedded impedance. Thus, let us discuss how to obtain Z_A by measurement. Two schemes come to mind.

The first and most direct is shown in Fig. D.12. Here all elements are fed via a harness as shown. The impedance Z_A of a typical element is determined by insertion of an appropriate reflectometer or impedance measuring device. Neither of these devices should have any attenuation, and their electrical length must be incorporated in the feed cable. From the complex reflection coefficient Γ at the element terminal we find

$$Z_A = Z_0 \frac{1 + \Gamma}{1 - \Gamma}. \quad (\text{D.11})$$

The second scheme is shown in Fig. D.13. Here all elements are loaded directly with the same load impedances Z_{L1} and exposed to an incident plane wave $\overline{E}^i$ (note: no connectors are necessary, strictly speaking). From the Thevenin equivalent circuit shown in the insert we then have

$$V_{L1} = V \frac{Z_{L1}}{Z_{L1} + Z_A}, \quad (\text{D.12})$$

where V is the induced generator voltage from the incident field (is independent of Z_L) and V_{L1} is the voltage across the load impedance Z_{L1} .

Similarly, for another load impedance Z_{L2} we have

$$V_{L2} = V \frac{Z_{L2}}{Z_{L2} + Z_A}. \quad (\text{D.13})$$

Division of (D.12) and (D.13) yields

$$\frac{Z_A + Z_{L2}}{Z_A + Z_{L1}} = \frac{Z_{L2}}{Z_{L1}} \frac{V_{L1}}{V_{L2}}. \quad (\text{D.14})$$

Knowing Z_{L1} and Z_{L2} , along with determination of the complex ratio V_{L1}/V_{L2} by measurements, enables us to find Z_A from (D.14).

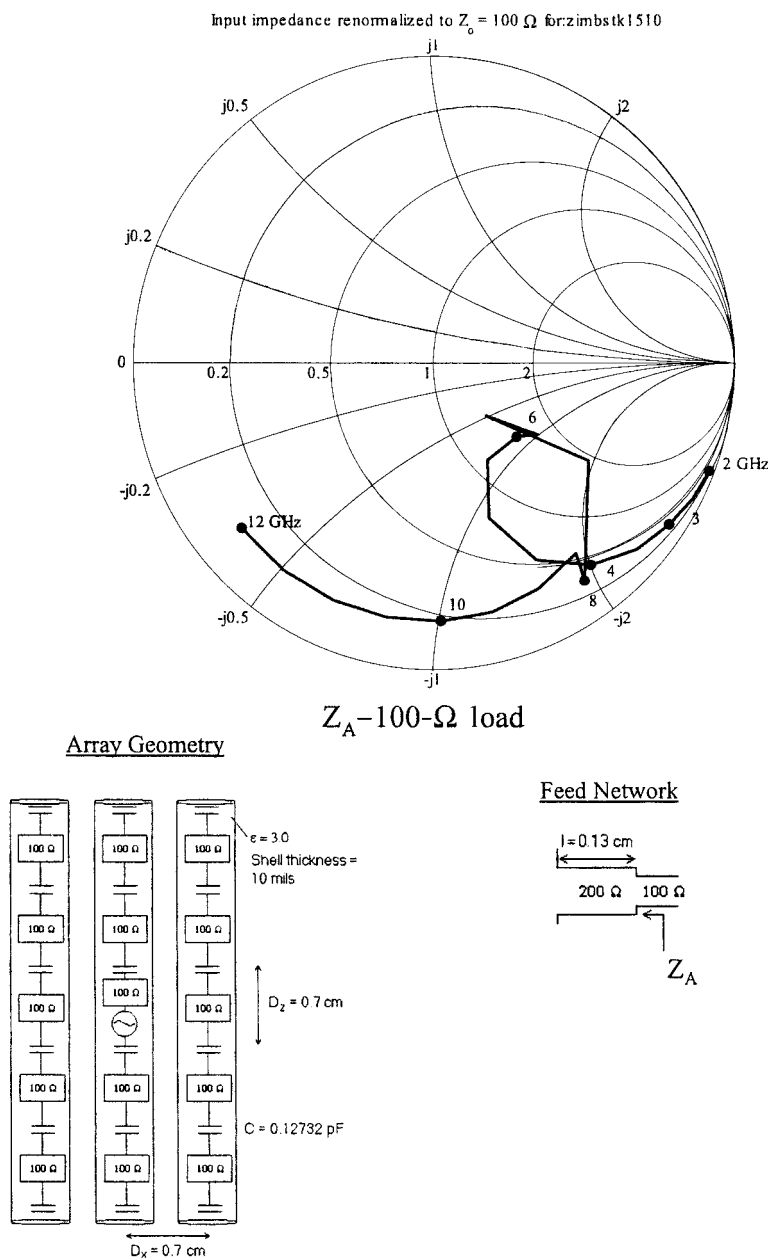


Fig. D.11 The impedance as seen from the center element with two parasitic sections at the top and two at the bottom like Fig. D.7b but flanked with parasitically excited columns of 5 segments as each side as shown in insert. From the ESP program; 10-mil dielectric. Includes matching sections (see text).

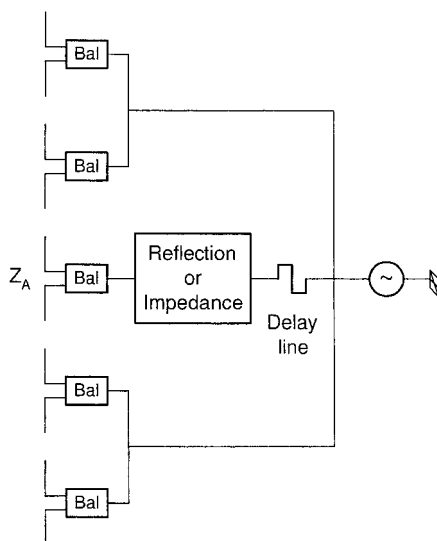


Fig. D.12 Determination of Z_A in a finite array by feeding all elements with the same voltages from “behind.” The individual impedance Z_A of a selected element is determined by an appropriate impedance measuring device as shown.

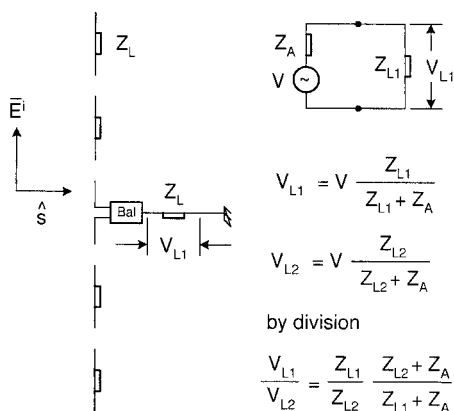


Fig. D.13 Determination of Z_A in a finite array by exposing all elements to the same incident field from the “front.” By knowing the load impedances Z_L in at least two cases and determination of V_L across the loads, we can determine Z_A (see insert).

The second approach does not require a harness or actual connectors as does the first approach. But we must use at least two different load impedances. Furthermore, measurement of the V_{L1}/V_{L2} takes sophisticated equipment. If any one out there “pulls it off,” the author would appreciate to hear from you. Congratulations would be in order.

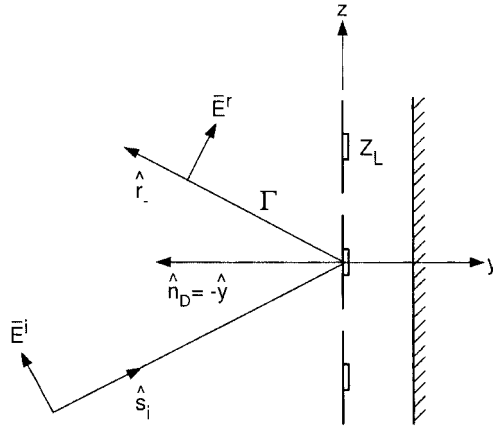


Fig. D.14 A plane wave incident upon an array of loaded dipoles in front of a groundplane.

Finally we show in Fig. D.14 a variation of the “frontal approach.” Here all elements in the array backed by a groundplane are simply terminated in the same load impedance Z_L (this requires no connectors). A plane wave $\vec{E}^i$ with direction of propagation $\hat{s}_i$ is incident upon this aperture. It is being reflected in the specular direction $\hat{r}_-$ with reflection constant Γ . The reflection coefficient observed from the terminals looking in the direction of $\hat{r}_-$ (or $\hat{s}_i$) has the same magnitude as Γ (not phase in general).

D.7 CONCLUSIONS

The investigation in this appendix clearly shows that the scan impedance and the embedded impedance are *not* the same, not even for broadside scan.

The fundamental reason for this is actually well known but practiced very little. In fact the embedded impedance is often demanded of the sponsors primarily for no other reason than “we have always done it that way in the past.”

If we merely plot the VSWR of the two impedances, they might exhibit some similarities, in particular for mediocre VSWR. This is by no means a proof of similarity. Impedance curves should preferably be plotted in the complex plane that always tells the complete story (and in particular what to do about matching; see Chapter 6 and Appendix B).

Thus, we have plotted all impedances in Smith charts normalized to the same impedance, namely 100 ohms. We can then compare the scan impedance Z_A as given by (D.5) in Fig. D.3 with the stick self-impedance $Z^{0.0}$ as shown in Fig. D.5. Although some similarities are found at the middle to higher frequencies, the difference at the lower frequencies is simply enormous. One could then wonder whether that difference could be reconciled if we instead of the stick self-impedance $Z^{0.0}$ would consider the embedded stick impedance $Z_{\text{emb stk}}$ as

given by (D.10) and depicted in Fig. D.6. That is obviously not the case. The reason is simply that $Z_{\text{emb stk}}$ according to (D.10) is the sum of $Z^{0,0}$ and two overcoupled terms from the parasitically excited arrays. These overcoupled terms reach their largest value when the outer stick self-impedances $Z^{-1,-1} + Z_L$ and $Z^{1,1} + Z_L$ including Z_L reach their minimum, namely around resonance at the middle frequencies. At the lower and higher frequencies, $Z^{-1,-1}$ and $Z^{1,1}$ are large, resulting in low values of the overcoupled impedances—that is, resulting in small deviations from $Z^{0,0}$.

To summarize: Using the embedded stick impedance $Z_{\text{emb stk}}$ instead of the stick self-impedance $Z^{0,0}$ may lead to even greater deviation from the scan impedance Z_A at the middle frequency range. At the lower frequencies, $Z^{0,0}$ and $Z_{\text{emb stk}}$ are similar but both differ substantially from the scan impedance Z_A .

An even greater difference is observed if we consider the embedded element stick impedance $Z_{\text{emb ele stk}}$ as shown in Fig. D.7. Here we excite only the center element of a single stick while all the other elements in the stick array are parasitically excited.

We finally demonstrated that a better approximation (in particular at the lower frequencies) to the scan impedance Z_A was obtained if we excited all columns in the finite array with voltage generators similar to the center columns and without any dummy elements whatsoever. To summarize, it seems that parasitic elements do more harm than good (at least for the type of array considered here).

D.8 POSTSCRIPT

Part of this appendix was written while I was recuperating at the Beth Israel hospital in Boston after some very delicate heart operations. I was consequently without my trusted associate Jonothan Pryor and his computer. Thus, I had to rely entirely on graphical approaches and my sliderule to verify my conclusions.

A typical example was to estimate the overcoupled impedance as given by the first and last term in (D.10). Making the rather gross approximation that

$$Z^{-1,1} = Z^{1,-1} \sim 0$$

reduces the overcoupled impedances in (D.10) to

$$-\frac{Z^{-1,0}Z^{0,-1}}{Z^{-1,-1} + Z_L} \quad \text{and} \quad -\frac{Z^{1,0}Z^{0,1}}{Z^{1,1} + Z_L}, \quad \text{respectively.} \quad (\text{D.15})$$

Further choosing $Z_L \sim 100$ ohms and making the rough approximation (at least at the middle frequencies) that

$$Z^{-1,-1} = Z^{1,1} \sim 100 + jC \frac{\Delta f}{f_0} \text{ ohms} \quad (C \text{ is a scaling constant}),$$

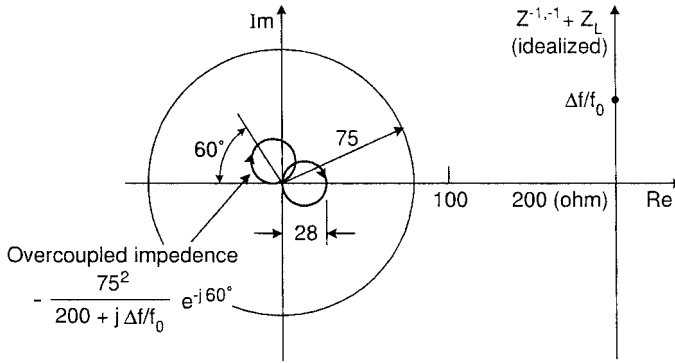


Fig. D.15 How to estimate the overcoupled impedance from the two outer stick arrays. See text for details.

we find that

$$Z^{-1,-1} + Z_L = Z^{1,1} + Z_L \sim 200 + jC \frac{\Delta f}{f_0} \quad \text{ohms}, \quad (\text{D.16})$$

that is, a straight vertical line through the point 200 ohm as shown in Fig. D.15. Furthermore, it was estimated that

$$Z^{-1,0} = Z^{0,-1} \sim 75e^{-j30^\circ}. \quad (\text{D.17})$$

Substituting (D.16) and (D.17) into (D.15) yields

$$-\frac{Z^{-1,0}Z^{0,-1}}{Z^{-1,-1} + Z_L} \sim -\frac{75^2}{200 + jC \frac{\Delta f}{f_0}} e^{-j60^\circ}. \quad (\text{D.18})$$

Inspection of (D.18) readily shows that it merely is an inversion of the vertical line going through 200 ohms into a circle with diameter $75^2/200 = 28$ ohms [123]. It is followed by the simple rotation $-e^{-j60^\circ}$. The inverted circles before and after rotation are also shown in Fig. D.15.

When the overcoupled impedances (remember we have two) are added to $Z^{0,0}$ as suggested by (D.10), we readily see that the embedded stick impedance $Z_{\text{emb stk}}$ will have a larger loop than the one present in $Z^{0,0}$ (see Fig. D.5). This is of course exactly what Jonothan later calculated (see Fig. D.6).

We may also conclude from (D.18) that a larger Z_L produces a smaller overcoupled impedance and vice versa.

Estimates as above are typically what I usually do before my students get the “right” values on the computer. Thus, I am ready to take a critical look at their results. It is just amazing to me when I think about the time and trouble some of them will go through to confuse and mislead a senile old professor.